READING
45 SOAL, 50 MENIT

Soal 1	Seagulls live on beach. They eat small fish, bread, and seaweed. Seagulls run quickly on the sand and fly quickly in the sky. Seagulls will run or fly away if you try to catch them. There are many seagulls on the beach. Crabs also live on the beach. They eat shrimp, ocean plants, and small fish. Crabs crawl quickly on the sand and in the ocean. Crabs will crawl away if you try to catch them. There are many crabs on the beach, but it is not always easy to see them. Starfish live on the beach, too. They eat clams, oysters, and small fish. Starfish move slowly on the sand and in the ocean. Starfish will not move away if you try to catch them. There are few starfish on the beach. Based on the information in the passage, which sentence is false? A. Seagulls can fly B. Crabs eat shrimp and ocean plants C. Seagulls move quickly on the sand and in the air D. There are a few starfish on the beach E. Starfish are hard to catch

Soal 2	Most of us heard the good advice that we need to eat less sugar – and rightly so. However, despite the numeros warnings by health authorities of the ill effects of sugar, the majority of the population is still consuming sugar on a daily basis in some from or other. We do not have to conusme white, refind sugar to be consuming sugar. Sugar includes glocuse, fructose (as in fruit sugar), lactose (as in milk), sucrose (as in table sugar), maltose or malts (as in rice malt and honey), jam (contains concentrated juice, which is high in fruit sugar), maple syrup, corn syrup, palm sugar (traditionally used in macrobiotic cooking), and the very deceiving organic brown sugar, which is not alot that different from white sugar. Even alcohol is a sugar. All of these sugars are problematic in many different ways. The body changes sugar into 2 to 5 times more fat in the bloodstream than it does starch. With 146 proven reasons why sugar is bad for us, is there perhaps one single reason as to why we might need it? The only interesting thing about sugar is that it tastes good and make us temporarily feel good. This is an area exploring. According to Traditional Chinese Medicine, a 5.000 years old wisdom of self-contained knowledge of healing, we all need sweetness in our life. We need six tastes: sweet, sour, slaty, astringent, bitter, and pungent to stimulate the taste buds on our tounge at main meals, in order to experience satiety. Many people really try hard to avoid sugar, and do not sweeten their tea or coffe, yet they crave sugar in some other from, such as chocolates, cakes, ice cream or even fruit – dates and figs. Dates are 99% sugar, in the from of fructose. When a person is in metabolic balance, they do noy crave sugar. If they do, it is a sign of metabolic imbalance and it can be corrected without having to consume sugar. The passage can be best summarized as ... A. all types of sugar are problematic to human health B. humans actually need sugar every day in the form of food C. given the ill effects of sugar, people should avoid consuming D. overconsumption of sugar will impact both physically and psychologically E. the only function of sugar is to satisfy our need of sweet taste

Soal 3	Jacob Davis bought large amounts of denim fabric from Levi Strauss. He was a tailor who made pants for hard-working men. One of his customers was continually tearing the pockets off his pants. So Jacob Davis decided to put rivets on certain parts of the pants to make them stronger. The customer loved the new pants so much that he told all his friends, and soon Jacob Davis was busy making lots of pants with rivets. Jacob Davis soon realized that using rivets was a great business idea, and he didn't want anybody to steal that idea. He decided that he would need to get a patent. But being a poor tailor, he didn't have enough money to pay for the patent. After thinking it over, he went to the businessman Levi Strauss and told him his idea. He said "If you agree to pay for the patent, we will share the profits from the riveted pants." Levi Strauss did agree, and the new riveted pant business was called Levi Strauss and Company. Today Levi's jeans are more popular than ever, and Levi's name continues to live on. What was Jacob Davis invented? A. Denim fabric B. Denim with pockets C. Denim pants D. Riveted denim E. A big company
Soal 4	Losari beach is a beautiful beach and located on the edge city of Makassar. It is located only about 3 km from the center of Makassar (Karebosi Park). The beach used to be the longest café in Asia, because many cafes stand in along the beach, but now the cafes are collected in a special place so it does not spread along the coast. Charm of the beach is mainly seen in the evening when the sunset stands out. This is a major attraction of people's coming to the Losari beach. Every evening hundreds of people come to witness the panorama of red as the sun will disappear into the ocean, so do not miss the sunset at the Losari beach. If the sky is sunny, the scenery is absolutely perfect. Because of its location in a bay, the water of Losari is even often quiet as usual pool water. Losari is its waterfront of Makassar. The lengthy of the beach is approximately one kilometer and it is a public space that can be accessed by anyone. On this beach there is a park called the Pelataran Bahari (Marine Park), with semicircular area of nearly one hectare. This place is a plaza with a clean floor for children to play and running around, while parents and teens sit on concrete benches to enjoy the sea breeze. From this place, you are also free to view out to the sea and watch the sunset slowly turns reddish in the line of the horizon. The reflected light also creates sheen on the surface of sea water. The Pelataran Bahari also serves as the stadium of open water to watch the coastal waters in front of Losari beach. This coastal water is often used as a racecourse jet

	ski, boat races and traditional boat jolloro katinting, or become a transit point of rely of Sandeq traditional sailboats and yachts. In Losari there are also a few hotels. Some of them qualified as a tree stars hotel. The hotel is offering panoramic beauty of the sea with luxury service treats. There are Losari Beach Hotel, Losari Beach Inn, Makassar Golden Hotel, and Pantai Gapura Hotel. All of the hotels located in Jalan Penghibur. Where can we watch the coastal water? A. In front of Losari Beach B. In Asia C. At the hotel D. In Makasar E. at Jalan Penghibur
Soal 5	Forget lions, tigers, and bears. When it comes to the art of war, army ants are among the most frightening creatures non earth. With powerful mouth parts, these fighters can skillfully cut creatures much larger than themselves into pieces. Acting together in great numbers, army ant colonies succeed at making tens of thousand of such kills each days. Their capabilities do have limits, though. Contrary to popular belief, they almost never take down large animals or people. One of the best places to observe army ants is Barro Colorado, an island in a lake created by the Panama Canal. The island is home to as many as 50 colonies of Eciton Burchellii, the most studied army ant in the world. It is one of 150 types of army ants in the New World. More than 170 other types live in Asia,Africa and Australia. The colonies of this army ant are huge, ranging from 300,000 to 700,000 ants. They never stay in one place long, moving from nest site to nest sit. Lingking legs together, they use their own bodies to from enormous nests called bivouacs, which they hang beneath a fallen tree. There they stay for about 20 days as the queen lays as many as 300,000 eggs. When the ants go hunting, as many as 200,000 of them leave the nest in a group that broadens into a fan as wide as 14 maters. This swarm raid takes a slightly different course each day, allowing the hunters to covers fresh ground each time. Protecting the ants wherever they go are the soldiers, recognizable by their overside jaws. If their frightening look don't scare enemies away, soldier also have a powerful bite and attack is often suicidal. Because their jaws are shaped like fishhook, the soldiers can't pull them out again. Amazonian tribes have used soldiers ants to close wounds, breaking off the bodies and leaving the head in place. *Eciton Burchellii* are blind and can't see what's ahead of them, but they move

	together in such great numbers that they easily kill the non-army ants, insects, and other small creatures that constitute their prey. When the groups *happen upon* a break in the path, ants immediately link legs togethers and from living bridges so that the groups can move forwards without any delay. In Japanese the word ant is written by linking two characters: one meaning "insect" the other meaning "loyalty". Indeed, individual ants are completely loyal to their fellow ants. They display many examples of selfless cooperation that, while certainly extreme, can't fail to win human admiration. The author's purpose of writing the text is to… A. Inform the readers about how army ants set up their life B. Demonstrate the strength of army ants protect their colonies C. Analyze how soldier ants protect their colonies D. Make the readers aware of the use of army ants E. Refute a common misconception about army ants
Soal 6	The most common causes of tsunamis are underwater earthquakes. To understand underwater earthquakes, you must first understand plate tectonics. The theory of plate tectonics suggest that the lithosphere, or top layer of the Earth, is made up of a series of huge plates. These plates make up the continents and seafloor. They rest on an underlying viscous layer called the asthenosphere. Think of a pie cut into eight slices. The pie crust would be the lithosphere and the hot, sticky, pie filling underneath would be the asthenosphere. On the earth, these plates are constantlyin motion, moving along each other at a speed of 1 to 2 inches (2.5-5 cm) per year. The movement occurs most dramatically along fault lines (where the pie is cut). These motions are capable of producing eartquakes and volcanism, which, when they occur at the bottom of the ocean, are two possible sources of tsunamis. When two plates come into contact at a region known as a plate boundary, a heavier plate can slip under a lighter one. This is called subduction. Underwater subduction often leaves enormous "handprints" in the form of deep ocean troches along the seafloor connected to the lighter plate may "snap up" suddenly due to pressure from the sinking plate. This result in a eathquakes. The focus of the earthquakes is the point within the Earth where the rupture first occurs, rocks break and the first seismic waves are generated, the epicenter is the point on the seafloor directly above the focus. When this piece of the plate snaps up and sends tons of rock shooting upward with tremendous force, the energy of that force is transferred to the water. The energy pushes the water upward above normal sea level. This is the birth os the tsunami. The eartquake that generated the December 26, 2004 tsunami in the Indian Ocean was 9.0 on the Richter scale – one of the biggest in recorded history. Which of the following natural disasters happening in Indonesia is related to the

	fault movement of plate tectonics? A. Volcanic eruption B. Soil erosion C. Forest fires D. Landslides E. Floods
Soal 7	Two state-owned lenders — Bank Mandiri and Bank BNI — will build temporary houses for 400 families displaced by the recent strong earthquake that hit Sigi District, CentralSulawesiProvinc Second Assistant Secretary of Sigi District Government Iskandar Nontji informed that the temporary houses for 100 families will be constructed by Bank Mandiri in the South Sibayala area of Dolo Sub-district. Meanwhile, Bank BNI will erect the temporary shelters for 300 families in Jonoge Village of Sigi Biromaru Sub-district that severely suffered from soil liquefaction during the 7.4-magnitude earthquake that rocked the area on Sept 28, he noted. The temporary houses will be constructed on pieces of land, owned by the sub-district government, village, and community members, he stated, adding that the Sigi district government will facilitate all related parties to sign a letter of agreement. “We are going to provide a letter of agreement that will be signed by all related parties to avoid any unexpected eventuality in the future,” he explained What makes 400 families displaced? A. Local governement B. 7.4-magnitude earthquake C. Temporary shelters D. 400 houses E. Mandiri bank and BNI

Soal 8	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text (1) The cause of tooth decay is acid, which is produced by bacteria in the mouth. The acid removes minerals from tooth enamel, allowing tooth decay to begin; the saliva in your mouth encourages remineralization and Line neutralizes the acid. The rate at which bacteria in the mouth produce acid depends on the amount of plaque on the teeth, the composition of the microbial flora, and whether the bacteria of the plaque have been "primed" by frequent exposure to sugar. To keep your teeth healthy, a regular dental hygiene program should be followed. (2) Removing plaque with a toothbrush and dental floss temporarily reduces the numbers of bacteria in the mouth and thus reduces tooth decay. **It** also makes the surfaces of the teeth more accessible, enabling saliva to neutralize acid and remineralize lesions. If fluoride is present in drinking water when teeth are forming, some fluoride is incorporated into the enamel of the teeth, making them more resistant to attack by acid. Fluoride toothpaste seems to acting another way, by promoting the remineralization of early carious lesions. (3) In addition to a regular dental hygiene program, a good way to keep your teeth healthy is to reduce your intake of sweet food. The least cavity- causing way to eat sweets is to have them with meals and not between. The number of times you eat sweets rather than the total amount determines how much harmful acid the bacteria in your saliva produce. But the amount of sweets influences the quality of your saliva. Avoid, if you can, sticky sweets that stay in your mouth a long time. Also try to brush and floss your teeth after eating sugary foods. Even rinsing your mouth with water is effective. Whenever possible, eat foods with fiber, such as raw carrot sticks, apples, celery sticks, etc., that **scrape off** plaque, acting as a toothbrush. Cavities can be greatly reduced if these rules are followed when eating sweets. We can infer from the passage that one benefit of fluoride to healthy teeth is …. A. It strengthens tooth enamel B. It stimulate saliva production C. It makes teeth whiter D. It is a replacement for brushing and flosing in dental care E. It produce a bacteria

Soal 9	Horse owners who plan to bread one or more mares should have a working knowledge of heredity and know how to care for breeding animals and foals. Then number of mares bred that actually concieve varies from about 40 to 85 percent, with the average running less than 50 percent. Some mares that do concieve fail to produce living foals. This means that, on average, two mares are kept a whole year to produce one foal, and even, some foals are disappointments from the standpoint of quality. By careful selection, breeders thoughout history have developed various kinds of horses with a wide variety of characteristics to suit many different needs. The Great Horse of the Middle Ages, for example, was bred for size amd strength to carry a heavily armored knight. The massive horses of such breeds are often called "cold blooded". The Arabs bred lithe desert horses that small and swift. These animals are often referred to as "hot blooded". "Cross-breeding of hot-blooded and cold-blooded horses" for certain characteristics produced breeds ranging from hiding horses to draft horses. The thoroghbred is considered by many to be the highpoint of elegance and fine selective breeding. Many persons mistakenly apply the name thoroughbred any purebred horse. Thoroughbred is a distinct breed of running horses that traces its ancestry through the male line directly back to four Eastern stallion; the Byerly Turk, the Darley Arabian,the Great horse and the Godolphin Barb. For convenience the breeds of horses are often divided into three major groups: (1) ponies, (2) heavy, or draft horses, and (3) light horses. In line 11 "little" most nearly means... A. Graceful B. Clumsy C. Massive D. Bulky E. Cute
Soal 10	Lucia Francisca Susi Susanti (born in Tasikmalaya, West java on 11 February 1971) was an Indonesian badminton player. Sometimes her nama is also spelled Susy Susanti. She won the Olympic Gold medal for woman's singles in the 1992 Olympic in Barcelona and the bronze medal at the 1996 Altanta Games. Susi Susanti is married to Alan Budi Kusuma, who also won a gold medal winner in 1992 Barcelona Olympic. Often considered as the finest female badminton player Indonesia ever produced, Susi Susanti dominated the women's singles event in the early 1990s, winning the All-England in 1990, 1991, 1993, and 1994, the World Grand Prix from 1990-1994, and the World Championships in 1993. Susi Susanti also led the Indonesian team to win the Uber Cup in 1994 and 1996. Susi Susanti was running a badminton racket company with her husband called Alan Susy Technology (Astec) after her retirement.

	"… Susi Susanti dominated the women's singles event in the early 1990s …" The antonym of the underlined word is … A. Manage B. Govern C. Rule D. Surrender E. Conquer
Soal 11	If you want farm green, two decisions about how you grow your crops are likely to have the biggest impact on achieving your goal, how much tillage you manage nitrogen. Although there probably are hundreds – or thousands – of other decisions, practices and technologies that can contribute to a green farming system, these two factors alone can come close to eliminating your cropping system's global warming footprint. At the same time, you will also reduce your operation's imapct on soil and water erosion, as well as surface and groundwater degradation from fertilizaers and other agricultural inputs. You may also be able to turn this environmental stewardship into extra cash by selling carbon credits associated with no-till. This surprising finding – that intensive crop production practices in use today can be nearly carbon neutral is emerging from research on both greenhouse gas production and greenhouse gas capture associated with production of annual crops. The research shows shows that basic crop management decisions to reduce tillage and use nitrogen more effeciently can cut a farming operation's impact on global warming gases 90% or more. In a best-case scenario, which wolud include no-till, these everyday practices come also to zeroing out a farming operation's greenhouse gas footprint. The author believes that tillage use and nitrogen management can help farmers.. A. reduce greenhouse gas production B. release carbon to the atmosphere C. increase shraphly D. use fertilizers more effectively E. decrease operational cost

Soal 12	Genesis is British progressive rock and band comprises of Phill Collins, Mike Rutherford, and Tony Banks (with touring members Chester Thompson and Daryl Stuermer) and previously including Peter Gabriel, Steve Hackett, Anthony Philips, and John Mayhew, and Ray Wilson, in various periods of their history. Genesis's first album, From Genesis to Revelation, was released in March 1969 and reached its international peak of #170 on the United States's Billboard 200 album chart. Over the next five years, Genesis released albums in the second half of every year, including two in 1973, as one was a live album. Over this period, the group's fame rose, as did their charting positions. Over time, the group has broken up, changed line ups, and taken both short and extended hiatuses. Which Genesis's album reached its international peak of #170 on the United States' Billboard 200 albums chart? *A. Nursery Cryme* *B. Trespass* *C. From Genesis to Revelation* *D. Selling England by the Pound* E. a live album
Soal 13	Cramming How do you study when the test is coming? Do you start preparing for the test weeks or months before the test or leave things to the last hour? If you start studying weeks or months before the test, it is great. However, if you study all the material in the last hour or minute, it is not good for you. This is called cramming. Cramming is when students stay up all night until morning to study before a test or finish an assignment. This habit can lead to negative impacts, the first being that disruptions in the regular sleep cycle can cause temporary intellectual lapses. For most students, less sleep can make them could not focus on the class. Additionally, cramming can leave us with memory lapses as well. Each person has a different sleeping schedule, so some of them often use a stimulant for cramming. An example stimulant, and the most common, is coffee. While delicious and beneficial, it causes many problems in the long-term such as caffeine intoxication syndrome, anxiety, panic, and headaches. To sum up, cramming is not recommended because it disturbs a person's sleep cycle which causes temporary intellectual lapses, and using stimulants for cramming gives them a bad effect on their health.

	Cramming is a method used by students when… A. they want to study B. they are going to face an exam C. they don't want to study D. they are too lazy E. they are diligent
Soal 14	Rainbows are formed when sunlight falling on raindrops is split into the different colours of the spectrum (range) of light. Sunlight is really a mixture of all the colours of the spectrum: red, orange, green, blue, indigo and violet, but our eyes always see them as ordinary white light. When a beam of light passes at an angle through a curved transparent surface, such as a raindrops, the beam is bent when it emerges. The different colours of light are bent by different amounts, so the white light is split into the colours of the spectrum. This effect can also be seen when light passes through a piece of glass cut at different angles, such as a prism. How are the rainbows formed? A. They are formed when sunlight falling on raindrops is split into the different colours of the spectrum of light B. When the sunlight spectrum mixed colours C. When a beam of light passes at an angle through a transparent surface D. When a beam of light passes through a piece of glass cut at different angles E. When raindrops and sunlight rise up together
Soal 15	Horse owners who plan to bread one or more mares should have a working knowledge of heredity and know how to care for breeding animals and foals. Then number of mares bred that actually concieve varies from about 40 to 85 percent, with the average running less than 50 percent. Some mares that do concieve fail to produce living foals. This means that, on average, two mares are kept a whole year to produce one foal, and even, some foals are disappointments from the standpoint of quality. By careful selection, breeders thoughout history have developed various kinds of horses with a wide variety of characteristics to suit many different needs. The Great Horse of the Middle Ages, for example, was bred for size amd strength to carry a heavily armored knight. The massive horses of such breeds are often called "cold blooded". The Arabs bred lithe desert horses that small and swift. These animals are often referred to as "hot blooded". "Cross-breeding of hot-blooded and cold-blooded horses" for certain characteristics produced breeds ranging from

	hiding horses to draft horses. The thoroghbred is considered by many to be the highpoint of elegance and fine selective breeding. Many persons mistakenly apply the name thoroughbred any purebred horse. Thoroughbred is a distinct breed of running horses that traces its ancestry through the male line directly back to four Eastern stallion; the Byerly Turk, the Darley Arabian,the Great horse and the Godolphin Barb. For convenience the breeds of horses are often divided into three major groups: (1) ponies, (2) heavy, or draft horses, and (3) light horses. The average amount of mares bred which actually concieve is less than what percent? A. 40 B. 85 C. 50 D. 75 E. 65
Soal 16	Holding on the belief that children will shed their “baby fat”as they get older may be perpetuating the childhood obesity epidemic, and laying a foundation for obesity later in life among overweight tots. According to 2006 data from the Center for Disease Control, 16% of American children were obese, and 32% overweight, with a growing propotion of overweight and obese children under age four. That is why it is critical to identify the “tipping point” between baby fat amd obesity risk, according to research from Eastern Virginia Medical School and Children’s Hospital of the King’s Daughter that was published this week in the journal Clinical Pediatrics. In a study of 111 overweight patients between the ages of two and 20 , researchers analyzed the patterns of children’s weight gain to determine at what age intervention to prevent obesity is most useful. Their conclusion: some children were on the road to obesity as early as thress months, meaning that doctors need to discuss unhealthy weight gain with parents as early as their child’s infancy, but certainly no later than age two. According to the passage, what is the danger in believing that baby fat will shed eventually? A. A growing proportion of overweight children B. It is perpetuating obesity epidemic among children C. Its critical to the tipping point between baby fat and obesity fat D. It is adding the pattern of children weight gain E. Parent will let their baby eats everything

Soal 17	What is photosynthesis? Photosynthesis is a food-making process and also produces oxygen that occurs in green plants. It is the chief function of leaves. The word photosynthesis means putting together with light. Green plants use energy from light to combine carbon dioxide and water to make sugar and other chemical compounds. How is the light used in photosynthesis? The light used in photosynthesis is absorbed by a green pigment called chlorophyll. Each food-making cell in a plant leaf contains chlorophyll in small bodies called chloroplasts. In chloroplast, light energy causes water drawn from the soil to split into hydrogen and oxygen. What are the steps of photosynthesis process? Let me tell you the process of photosynthesis, in a series of complicated steps, the hydrogen combines with carbon dioxide from the air, forming a simple sugar. Oxygen from the water molecules is given off in the process. From sugar together with nitrogen, sulphur, and phosporus from the soil-green plants can make starch, fat, protein, vitamins, and other complex compounds essential for life. Photosynthesis provides the chemical energy that is needed to produced these compounds. What the product of photosynthesis ... A. Sugar B. Food and O2 C. Root D. Food E. Branch
Soal 18	Calorie-restricted diets have been shown to increase the lifespan and health span of everything from yeast to monkeys – as long as there is no malnutrition. And while no long-term studies have proven the benefits of calorie restriction on human lifespan, shorter-term studies suggest that it does improve health. Here's how it might work. Our bodies monitor and sense the amount of nutrients available through specific molecules in our cells. Depending on the amount of food we eat, these molecules tweak our metabolism to regulate how we use the available nutrients. One of these molecules is an enzyme called TOR. When there is a lot of food, the TOR enzyme instructs cells in the body to grow. If there is less food, TOR instructs the body to be on alert - a state that scientists refer to as a "mild stress response". Many experiments have shown that when animals eat a lot of food, especially for prolonged periods, TOR senses this and their lifespan becomes shorter. But do all foods have this effect on TOR? TOR enzyme is especially activated when cells sense large amounts of amino acids (the building blocks of protein) or protein. A protein-restricted diet, without malnutrition, can have the same effects on the

	metabolism and lifespan of lab animals as a calorie-restricted diet. Age-related diseases are known to be caused by genetic mutations, but could there be a connection between TOR, nutrition and diseases of old age? We know that nutrition is associated with cancer and heart disease, and overactive TOR is known to be involved in these diseases, but recent studies show that TOR is also directly related to neurodegenerative diseases. For example, the activity of the TOR enzyme in the brains of people with Alzheimer's is much higher compared with healthy brains. Also, simulating these diseases in mice and other lab animals has shown that removing excess TOR stops brain cells dying. So, there may be a link between what we eat, how it is sensed by our body and the risk of neurodegenerative disease. Scientists are exploring different possibilities to prevent neurodegeneration. If more protein means more active TOR, we could either modify our diet, safely, or develop a drug that tricks our body into thinking it is getting less protein. source: https://theconversation.com/tor-an-enzyme-that-could-hold-the-secret-to-longevity-and-healthy-ageing-118568 According to the text, how can the risk of neurodegenerative diseases be reduced? A. by managing our protein and dietary intake B. by linking what we eat and how it is sensed C. by including high-protein foods in our diet D. by consuming medications that tricks our body E. by enhancing TOR enzyme activity in the body
Soal 19	Wetlands are areas of land that are covered by water, or have flooded or waterlogged soils. They can have water on them either permanently or for just part of the year. Whether it's year-round or seasonal, this period of water saturation produces hydric soils, which contain little or no oxygen. But this doesn't mean that they are lifeless: Wetlands are full of unique water-loving plants and wildlife that have adapted to wet environments. Wetlands can take many different forms, depending on the local climate, water conditions and land forms and features. For example, swamps are dominated by woody trees or shrubs. Marshes often have more grass-like plants, such as cattails and bulrushes. Bogs and fens are areas that accumulate peat – deposits of dead and partly decomposed plant materials that form organic-rich soil. Wetlands are important environments for many reasons. They provide ecological services whose value has been estimated to be worth more than US$47 trillion per year. For example, wetlands support very high levels of biodiversity. Scientists estimate that 40% of all species on Earth live or breed in wetlands. Wetlands are critical homes or stopovers for many species of migratory birds. In the central U.S. and Canada, for example, wetlands in the so-called prairie pothole region on the Great Plains support up to three-quarters of North America's breeding ducks.

	Along with providing important habitat for everything from microbes to frogs to waterfowl, wetlands also work to improve water quality. They can capture surface runoff from cities and farmlands and work as natural water filters, trapping excess nutrients that otherwise might create dead zones in lakes and bays. Wetlands can also help remove other pollutants and trap suspended sediments that cloud water bodies, which can kill aquatic plants and animals. Because wetlands are often in low-lying areas of the landscape, they can store and slowly release surface water. Wetlands can be extremely important for reducing the impacts of flooding. In some places, water entering wetlands can also recharge groundwater aquifers that are important for irrigation and drinking water. Wetlands also act as important carbon sinks. As wetland plants grow, they remove carbon dioxide from the atmosphere. They die, sink to the bottom of the wetland and decompose very slowly. Over time, the carbon they contain accumulates in wetland soils, where it can be stored for hundreds of years. Conserving and restoring wetlands is an important strategy for regulating greenhouse gases and mitigating the impacts of climate change. source: https://theconversation.com/what-is-a-wetland-an-ecologist-explains-191495 According to the passage, all of the following are true about wetlands EXCEPT…. A. they are regions of land that are submerged in water B. they can appear in a wide variety of shapes and forms C. they may be covered with water periodically or all year D. they are completely inhospitable to any form of life E. they generate hydric soils with little or no oxygen
Soal 20	Wind energy is produced by the movement of air and converted into power for human use. Wind has been used as a source of energy for more than a thousand years, but was replaced by fossil fuels for much of the 20th century. Today, wind is making a comeback as a source of electricity and power. Wind energy is produced with wind turbines—tall, tubular towers with blades rotating at the top. When the wind turns the blades, the blades turn a generator and create electricity. Wind turbines can have a horizontal or vertical axis. The turbines do not actually produce wind energy. The blades turn, convert the energy of wind into rotational energy, a form of mechanical energy, and this energy is in turn converted into electrical energy. Horizontal-axis wind turbines (HAWTs) are the most familiar type of electricity-producing windmill. Most have three large blades that spin parallel to their towers, where the main rotor and generator are located. Most HAWT arrays are painted white, to promote visibility to low-flying aircraft. This type of turbines can be much bigger than their vertical counterparts.

	Vertical-axis wind turbines (VAWTs) have varied, unusually shaped blades that rotate in complete circles around their tower. The main rotor and generator are located near the ground, making maintenance easier and less expensive. VAWTs do not have to be upwind to generate electricity. Vertical-axis wind turbines can be much smaller than their horizontal counterparts. Standing only 5 meters (15 feet) tall, these VAWTs can be installed on the roofs of buildings. Turbines cannot operate at every wind speed. If winds are too strong, they can be damaged. Therefore, the turbine has an automatic controller that turns on when winds are blowing at prime speed for generating electricity. This speed is usually 13 to 90 kilometers per hour (8 to 55 miles per hour). If the winds become stronger than that, the controller turns the turbine off. source: https://education.nationalgeographic.org/resource/putting-wind-work Which of the following is not the advantages of VAWT turbines mentioned in the passage? A. They are easy to repair B. They occupy less room C. They can operate in any conditions D. They require lower maintenance cost E. They can work in any wind direction
Soal 21	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text Healthy eating is not about strict dietary limitations, staying unrealistically thin, or depriving yourself of the foods you love. Rather, it is about feeling great, having more energy, improving your outlook, and stabilizing your mood. If you feel overwhelmed by all the conflicting nutrition and diet advice out there, you are not alone. It seems that for every expert who tells you a certain food is good for you, you will find another saying exactly the opposite. But by using these simple tips, you can cut through the confusion and learn how to create a tasty, varied, and healthy diet that is as good for your mind as it is for your body. We all know that eating right can help maintain a healthy weight and avoid certain health problems, but your diet can also have a profound effect on your mood and sense of well being. Studies have linked eating a typical western diet filled with processed meats, packaged meals, takeout food, and sugary snacks - with higher rates of depression, stress, bipolar disorder, and anxiety. Eating an unhealthy diet may even play a role in the development of mental health disorder such as ADHD, Alzheimer's disease, and schizophrenia, or in the increased risk of suicide in young people.

	Eating more fresh fruits and vegetables, cooking meals at home, and reducing your intake of sugar and refined carbohydrates, on the other hand, may help to improve mood and lower your risk for mental health problems. If you have already been diagnosed with a mental health problem, eating well can even help to manage your symptoms and regain control of your life. While some specific foods or nutrients have been shown to have a beneficial effect on mood, it is your overall dietary pattern that is most important. That means switching to a healthy diet does not have to be an all or nothing proposition. You do not have to be perfect, and you do not have to completely eliminate foods you enjoy to have a healthy diet and make a difference to the way you think and feel. Based on the passage, it can be hypothesized that a good diet will impact the body if .... A. The dietary menu is under a nutrition expert control B. The meals are home-cooked in traditional fashion C. What is eaten equalizes the feeling of en- joying the food D. A good selection of fruit and vegetable becomes the menu E. Diet is losing controlled for minimizing distress and stress
Soal 22	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text John Apollos is losing weight the old-fashioned way - by eating less. A whole lot less. As a volunteer in the two year Comprehensive Assessment of long-Term Effects of Reducing Intake of Energy (CALERIE) study at Tufts University in Boston, Apollos has lowered his daily calorie intake 25% over the past eight months. The fat, not surprisingly, has melted away; the 52-year-old physical trainer has lost more than 11 kg since the study began and is down to his high school weight. Yet, that's not the real reason Apollos and the other participants in the program are eating only three quarters of what they used to. The researchers running the multicenter CALORIE study are trying to determine whether restricting food intake can slow the aging process and extend our life span. "I feel better and lighter and healthier," says Apollos. "But if it could help you live longer, that would be pretty amazing. The idea is counterintuitive: If we eat to live, how can starving ourselves add years to our lives? Yet, decades of calories restriction studies involving organisms ranging from microscopic yeast to rats have shown just that, extending the life spans of the semi starved as much as 50%. Last July a long-term study led by researchers at the University of Wisconsin nudged the implications of this a bit closer to our species, finding that calorie restriction seemed to extend the lives of humanlike rhesus monkey as well. The hungry primates fell victim to diabetes, heart and brain disease and cancer much less frequently than their well-fed counterparts did.

	However, there may be more than just the absence of disease operating here. Anytime you go on a diet, after all, you stand a good chance of lowering your blood pressure, cholesterol level and risk of diabetes and other health woes. All that can translate into extra years. With calorie restriction, usually defined as a diet with 25% to 30% fewer calories than normal but still containing essential nutrients, something else appears to be at work to extend longevity. Which of the following ideas from the text above contains an opinion? A. Restricting calorie consumption for longevity. B. Absence of disease due to controlling calorie intake. C. Apollos' feeling better and lighter and healthier. D. The melting of body fat by having less foods. E. Apollos' restricting his daily calorie consumption.
Soal 23	Calorie-restricted diets have been shown to increase the lifespan and health span of everything from yeast to monkeys – as long as there is no malnutrition. And while no long-term studies have proven the benefits of calorie restriction on human lifespan, shorter-term studies suggest that it does improve health. Here's how it might work. Our bodies monitor and sense the amount of nutrients available through specific molecules in our cells. Depending on the amount of food we eat, these molecules tweak our metabolism to regulate how we use the available nutrients. One of these molecules is an enzyme called TOR. When there is a lot of food, the TOR enzyme instructs cells in the body to grow. If there is less food, TOR instructs the body to be on alert - a state that scientists refer to as a "mild stress response". Many experiments have shown that when animals eat a lot of food, especially for prolonged periods, TOR senses this and their lifespan becomes shorter. But do all foods have this effect on TOR? TOR enzyme is especially activated when cells sense large amounts of amino acids (the building blocks of protein) or protein. A protein-restricted diet, without malnutrition, can have the same effects on the metabolism and lifespan of lab animals as a calorie-restricted diet. Age-related diseases are known to be caused by genetic mutations, but could there be a connection between TOR, nutrition and diseases of old age? We know that nutrition is associated with cancer and heart disease, and overactive TOR is known to be involved in these diseases, but recent studies show that TOR is also directly related to neurodegenerative diseases. For example, the activity of the TOR enzyme in the brains of people with Alzheimer's is much higher compared with healthy brains. Also, simulating these diseases in mice and other lab animals has shown that removing excess TOR stops brain cells dying. So, there may be a link between what we eat, how it is sensed by our body and the risk of neurodegenerative disease. Scientists are exploring different possibilities to prevent neurodegeneration. If more protein means more active TOR, we could

	either modify our diet, safely, or develop a drug that tricks our body into thinking it is getting less protein. source: https://theconversation.com/tor-an-enzyme-that-could-hold-the-secret-to-longevity-and-healthy-ageing-118568 According to the text, what is the role of TOR enzyme? A. It alters nutrition regulation and human metabolism B. It monitors available nutrients and digests them C. It produces cells and activate mild stress response D. It control body molecules and prolonged lifespan E. It detects food intake and regulates cell growth
Soal 24	Cement isn't something most people notice, even though we all depend on it every day. The most common type of cement is called Portland cement, or OPC (ordinary Portland cement). Portland cement is used to bind concrete, an essential component of roads, bridges, and buildings. Cement production is also a big contributor to global warming. Manufacturing Portland cement requires heating limestone to 1,449 degrees Celsius (2,640 degrees Fahrenheit). The heating process uses fossil fuels, which release massive amounts of carbon dioxide into the atmosphere. Carbon build-up in the Earth's atmosphere causes global warming. A company called Calera believes it has figured out a way to make cement that stores carbon instead of releasing it into the atmosphere. Calera combines flue gas released from power plants with seawater to make cement. This method of carbon storage is called carbon sequestration. Today, Calera has a prototype plant in Moss Landing, on the coast of Monterey Bay, California. Flue gas from the natural gas facility is pumped into Calera's cement plant. The flue gas, filled with carbon dioxide, is combined with seawater from nearby Monterey Bay. Seawater contains calcium, magnesium, and other minerals, the same ingredients corals use to build reefs. Carbon dioxide from the flue gas and minerals in the seawater combine to make calcium carbonate. Calcium carbonate is the material that makes up coral reefs, as well as Portland cement. The process of making calcium carbonate cement from carbon dioxide and seawater is called Mineralization via Aqueous Precipitation, or MAP. After the cement is made, the heat from the flue gas is used to dry the cement into powder. Calera plans to desalinate and sell the leftover seawater. The idea for MAP was developed by Brent Constantz, one of the co-founders of Calera. As a graduate student at the University of California at Santa Cruz, Constantz observed how corals used carbon dioxide to make reefs. He thought that if corals could make cement this way, then perhaps humans could do it, too.

	source: https://education.nationalgeographic.org/resource/working-with-nature-to-slow-global-warming According to the passage, why is cement production a significant contributor to global warming? A. Because it raises the temperature of the limestones to 1,449 degrees Celsius B. Because it is used to bind concrete in the construction of most buildings C. Because it is necessary for activities that people engage in daily basis D. Because it uses fossil fuels which emit carbon dioxide into the atmosphere E. Because it stores carbon dioxide releases in the processing of limestone heating
Soal 25	Commonly assumed to be silent, 53 animals have had their 'voices' added to a family tree of vocalizations in an effort to determine when acoustic communication emerged in evolutionary history. The species that are finally being heard come from four different animal clades, including 50 turtle species, the South American lungfish (Lepidosiren paradoxa), a limbless amphibian called Cayenne caecilian (Typhlonectes compressicauda), and a reptile from New Zealand known as a tuatara (Sphenodon punctatus). All recorded species were found to possess a varied acoustic repertoire comprising a number of different sounds. The fact that so many species across the turtle tree of life share an ability to make vocal sounds strongly suggests it is ancestral to the entire turtle clade. What's more, when researchers combined the turtle data with an analysis of 1,800 other vertebrates with lungs, they found evidence of a common origin of sound production and acoustic communication across all animals in possession of enough backbone to draw and exhale breath. Amphibians and reptiles are much less studied, but just because the researchers don't have much data on their sound production, doesn't mean they aren't making noise. If we really want to understand how acoustic communication evolved, researchers argue, we have to systematically document these "neglected groups". This is exactly what the current study attempted to test. The authors only searched for sound production among 106 neglected species. Today, the reigning theory on acoustic communication suggests that this fundamental trait arose multiple times across the animal tree of life. This is based on an apparently wide variety of ear structures and vocal morphologies seen among birds and mammals. But this new phylogenetic analysis suggests that is incorrect. The fact that animals so far apart on the tree of life are producing sound in similar ways and for similar reasons, whether it be for mating, communication, or parenting, suggests the skill shares a common origin. It might have evolved just once before diverging. In fact, the trait seems to date back to an ancestor common to a wide variety of modern vertebrates

	that lived 407 million years ago. "Our results now show that acoustic communication did not evolve multiple times in diverse clades, but has a common and ancient evolutionary origin," concludes paleobiologist Marcelo Sánchez from the University of Zurich in Switzerland. The findings are a good reminder that just because an animal is hard to hear, doesn't mean we shouldn't listen to them. source: https://www.sciencealert.com/we-thought-these-animals-were-silent-scientists-just-found-their-voices According to the text, how does this new study refute the prior theory of how animal talks? A. It support the claims that animal can communicate multiple times in their lifetime B. It argues that only certain kinds of animals in the tree of life can produce sound C. It contradicts that animals making sounds in similar ways and for similar reasons D. It claims that the phylogenetic analysis is incorrect description of how animal talks
Soal 26	The term plantation arose as settlements in the southern United States, originally linked with colonial expansion, came to revolve around the production of agriculture. The word plantation first appeared in English in the 15th century. Originally, the word meant to plant. However, what came to be known as plantations became the center of large-scale enslaved labor operations in the Western Hemisphere. Historians Peter H. Wood and Edward Baptist advocate to stop using the word plantation when referencing agricultural operations involving forced labor. Instead they suggest calling these places “labor camps” or “slave labor camps.” The plantation system developed in the American South as British colonists arrived in what became known as Virginia and divided the land into large areas suitable for farming. The land on which these plantations were established was stolen through canceled, disregarded, and deceitful treaties, or outright violence towards indigenous nations. The plantation system came to dominate the culture of the South, and it was rife with inequity from the time it was established. In 1606, King James I formed the Virginia Company of London to establish colonies in North America, but when the British arrived, they faced a harsh and foreboding wilderness, and their lives became little more than a struggle for survival. So, to make settling the land more attractive, the Virginia Company offered any adult man with the means to travel to America 50 acres of land. At the encouragement of the Company, many of the settlers banded together and created large settlements, called hundreds, as they were intended to support 100 individuals, usually men who led a household. The hundreds were run as private plantations intent on making a profit from the

	cultivation of crops, which the economy of the South depended on. The climate of the South was ideally suited to the cultivation of cash crops. Unlike small, subsistence farms, plantations were created to grow cash crops for sale on the market. The plantation system was an early capitalist venture. England's King James had every intention of profiting from plantations. Tobacco and cotton proved to be exceptionally profitable. source: https://education.nationalgeographic.org/resource/plantation-system According to the passage, how did the British colonist obtain plantation lands in South America? A. The British negotiated with the landowners B. The indigenous nations canceled the deals C. The invaders established deceitful treaties D. The colonist divided the areas for farming E. The natives were robbed of these lands
Soal 27	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text (1) High school students may improve their science grades by learning about the personal struggles and failed experiments of great scientist such as Albert Einstein and Marie Curie. (2) In one study, 402 students from New York City high schools in low-income areas of the Bronx and Harlem were divided into three groups. The control group read an 800-word typical science and textbook description about the great accomplishments of Einstein, Curie and Michael Faraday. **(3)** Another group read about those scientists' personal struggles, including Einstein's flight from Nazi Germany to avoid persecution as a Jew. The third group of students read about the scientists' intellectual struggles, such as Curie's persistence despite a string of failed experiments. The struggle stories included actions the scientist took to overcome these **hurdles.** (4) Students who read about the scientist' intellectual or personal struggles were more likely to say the famous scientist were people, like themselves, who had to overcome failure and obstacles to succeed. Students in the control group more often believed the great scientist had innate talent and a special aptitude for science. (5) The study suggests that science textbooks should highlight the struggles of great scientists and provide more vivid narrative descriptions of the techniques that scientists used to overcome challenges. "Many kids do not see science as part of their everyday lives. We teach them important content, but we never bring it to

	life," said the lead researcher Xiaodong Lin-Siegler. PhD. "Our science curriculum is impersonal, and kids have a hard time relating to it because they just see a long list of fact that they have to memorize. What is the suggestion of the study? A. Scientists have to be like ordinary people. B. Scientists had to overcome failure to succeed. C. Great scientists had an innate talent and a special aptitude for science. D. Science textbooks should emphasize the struggles of great scientists. E. Science textbooks should provide more descriptions of great scientists' success.
Soal 28	Dear Melissa, I hope you are well. Guess what! I am having a big party next week at my house and I am inviting all my close friends. The theme of the party is going to be Harry Potter, so all must dress accordingly. You would make a great witch! I need to let everyone know by Wednesday since I am throwing the party next Saturday. Sam is going to decorate the house and my mother is going make a cake and make lots of food. There is going to be music and dancing as well. John is going to be the DJ. You can stay over at my house. It's going to be great! I hope to see you then. Love, Sarah What kind of an organization is it? A. It is a slumber party B. It is a costume party C. It is about making a cake D. It is a dance party E. It is a music party
Soal 29	Wetlands are areas of land that are covered by water, or have flooded or waterlogged soils. They can have water on them either permanently or for just part of the year. Whether it's year-round or seasonal, this period of water saturation produces hydric soils, which contain little or no oxygen. But this doesn't mean that they are lifeless: Wetlands are full of unique water-loving plants and wildlife that have adapted to wet environments. Wetlands can take many different forms, depending on the local climate, water conditions and land forms and features. For example, swamps are dominated by woody trees or shrubs. Marshes often have more grass-like plants, such as cattails and bulrushes. Bogs and fens are areas that accumulate peat – deposits of dead and

	partly decomposed plant materials that form organic-rich soil. Wetlands are important environments for many reasons. They provide ecological services whose value has been estimated to be worth more than US$47 trillion per year. For example, wetlands support very high levels of biodiversity. Scientists estimate that 40% of all species on Earth live or breed in wetlands. Wetlands are critical homes or stopovers for many species of migratory birds. In the central U.S. and Canada, for example, wetlands in the so-called prairie pothole region on the Great Plains support up to three-quarters of North America's breeding ducks. Along with providing important habitat for everything from microbes to frogs to waterfowl, wetlands also work to improve water quality. They can capture surface runoff from cities and farmlands and work as natural water filters, trapping excess nutrients that otherwise might create dead zones in lakes and bays. Wetlands can also help remove other pollutants and trap suspended sediments that cloud water bodies, which can kill aquatic plants and animals. Because wetlands are often in low-lying areas of the landscape, they can store and slowly release surface water. Wetlands can be extremely important for reducing the impacts of flooding. In some places, water entering wetlands can also recharge groundwater aquifers that are important for irrigation and drinking water. Wetlands also act as important carbon sinks. As wetland plants grow, they remove carbon dioxide from the atmosphere. They die, sink to the bottom of the wetland and decompose very slowly. Over time, the carbon they contain accumulates in wetland soils, where it can be stored for hundreds of years. Conserving and restoring wetlands is an important strategy for regulating greenhouse gases and mitigating the impacts of climate change. source: https://theconversation.com/what-is-a-wetland-an-ecologist-explains-191495 Which of the following is NOT mentioned in the passage as the benefits of wetlands? A. Essential animal habitat B. Water purification C. Carbon dispersion D. Mitigate flooding E. Biodiversity protection
Soal 30	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text (1) The Hawaiian Islands are a tropical paradise most of the time. But even there, nature can sometimes show its more violent side. The islands have been bit by

	numerous natural disasters in the recent past. These disasters have caused deaths and great damage. (2) Through all of Hawaii's history, volcanoes have played an important part. The islands were formed by the eruption of volcanoes under the ocean. In the big island of Hawaii, the volcanoes continue to erupt. During an eruption, hot lava pours out of the top of a volcano and down the side. The red, hot lava covers everything in its path. The air is filled with smoke and horrible smelling gas that is sometimes poisonous. (3) Native Hawaiians believe that volcanoes are a natural part of life. In their traditional religion, volcanoes are ruled by the goddess, Pele. When Pele is angry, the volcanoes erupt. Before an eruption she takes a human from. If you meet her and are kind to her, you will not be hurt by the lava. Today, Hawaiians have forgotten most of the ancient gods, but not Pele. They continue to honor her and her volcanoes. (4) Another kind of natural disaster that has hit Hawaii is the tidal wave, or tsunami. Unlike volcanoes, tsunamis come from far away under the ocean. When an earthquake occurs underwater, it creates huge waves. If these waves hit land, they destroy everything along the coast. (5) These days, all of the Hawaiian Islands have special signals to warn of tsunamis. When the signal sounds, everyone knows it is time to run. They must try to get as far as from the seashore as possible. It can be inferred from the passage that most Hawaiians today .... A. Follow their traditional religion B. Do not follow their traditional religion C. Do not understand volcanoes D. Do not worry about storms E. Do not worry about tsunamis
Soal 31	Scientists have found fossils of animals with tails dating back hundreds of millions of years. Back then, early fish used their fanlike tails as fins to swim through oceans and escape predators. As these fish evolved into creatures that lived on the land, their tails started to change too. Whether they belong to reptiles, insects, birds or mammals, tails serve a wide variety of purposes. Modern animals use their tails for everything from balance to communication and finding mates. Scientists believe that dinosaurs, including the Tyrannosaurus rex, swung their tails side to side to balance their heavy heads and bodies while walking on two legs. This movement allowed them to run fast enough to catch their prey. Similarly, present-day kangaroos use their tail for balance when they leap across the open land. But they don't just use it as a counterbalance for their weight – the kangaroo's tail also functions as a powerful third leg that can help propel them through the air.

	Cats and other animals that climb often have bushy or long tails that help them balance, kind of like a tightrope walker holding a long pole. Monkeys use their long tail for balance while swinging through forest tree branches. Many have prehensile, or grasping, tails that act like hands and allow them to hold onto tree limbs. These tails are so strong that they can even hold the animal up while it eats fruit and leaves. Other animals' tails evolved into weapons. For instance, stingrays have a trademark stinger tail they can use as a defense when a predator attacks them. Venomous rattlesnakes have buttons of dried skin on their tail that make a racket when they shake it. This warns any animals that might threaten the rattlesnake that it's getting ready to strike. Many insects also have tails, but they evolved separately from other animals with backbones, like fish and mammals. Most tailed insects use their tails to lay eggs or to sting and paralyze hosts or prey. In some animals, like wasps, their tails can do both, as certain parasitic wasps will lay their eggs inside a host. In the other hand, grazing animals, use their tails to swat off insects that may be bothering them. source: https://theconversation.com/why-do-animals-have-tails-187575 Which of the following can be inferred about creatures which live on the land mentioned in paragraph 1? A. They lived both in the ocean and land B. They used the tails for self-defense C. They were previously aquatic creatures D. Their tails were changed into fins E. Their tails were shaped like fins
Soal 32	Archeologists have long speculated that Stonehenge, the prehistoric stone circle in Wiltshire, England, was a sort of astronomical calendar because of its perfect alignment with the midsummer and midwinter solstices. However, they have struggled to determine how the calendar worked. Now, Professor Timothy Darvill of Bournemouth University in the UK, may have finally solved the mystery. The expert believes the 5,000-year-old neolithic monument served as a solar calendar for the local residents. "It's a perpetual calendar that recalibrates every winter solstice sunset," Darvill says. "This would have enabled the ancient people who lived near the monument in what is now Wiltshire, UK, to keep track of days and months of the year." Darvill's quest to find the monument's true purpose began in 2020 after a new study revealed that most of the sarsen stones — the rocks found at Stonehenge — were from the same local quarry. They were also taken and set up at Stonehenge at the same time. The expert says this indicated they served a common cause.

	To find out what that could be, Darvill turned to the monument's arrangement for clues. Stonehenge's outer circle originally comprised 30 large sarsen stones linked together by 30 stone crossbars, or lintels. Darvill believes they represented the 30 days of the month. The researcher further theorizes that the five sets of trilithons — pairs of large vertical stones each with a horizontal stone atop — that lie within the Sarsen Circle, represent five additional days. The archeologist further asserts that the four so-called "station stones" outside the sarsen circle served as a reminder to add a leap day every four years. "Thirty, 5, and 4 are interesting numbers in a calendrical kind of sense," says Darvill. "Those 30 uprights around the main sarsen ring at Stonehenge would fit very nicely as days of the month. Multiply that by 12, and you get 360, add on another 5 from the central trilithons, you get 365." Darvill, who published his findings in the journal Antiquities on March 2, 2022, is unsure why the ancient people decided to develop a solar calendar. He thinks they may have gotten the idea from the Mesopotamians and Egyptians, who also created solar calendars during that time. It is also possible that they embarked on the initiative on their own. source: *https://www.dogonews.com/2022/3/23/stonehenge-may-have-served-as-a-solar-calendar* What can be inferred from the sixth paragraph about the Stonehenge ancient people? A. They are more advanced than Mesopotamians and Egyptians B. They initiated the construction of Stonehenge on their own C. They presumably influence by other civilization at the period D. It is still unclear why they decided to build a solar calendar E. It is decided that they created solar calendars during that time
Soal 33	Photosynthesis is the prcess of converting solar energy to chemical energy and storing it bonds of sugar. This process occurs in plants and some algae (Kingdom Protista). Plants need only solar energy, CO2 and H20 to make sugar. The process of photosynthesis takes in the chloroplasts, specifically using chlorophyll, the green pigment involved in photosynthesis. Photosynthesis takes place primarily in plant leaves, and little to none occurs in stern, etc. The part of typical leaf include the upper and lower epidermis , the mesophyll, the vascular bundle(s) (venis), and the stomates. The upper and lower epidermal cells do not have chloropalsts, thus photosynthesis does not occurs there. They serve primarily as protection for the rest of the leaf. He stomates are holes which occur primarily in the lower epidermis and are for air exchange : they let CO2 in and O2 out. The vascular bundles or veins in a leaf are part of the plant’s transportation system, moving water and nutrient around the plant as needed. The mesophyll cells have chloroplasts and this is where photosynthesis

	occurs. As you hopefully recall, the parts of chloroplast include the outer and inner membranes, intermembrane space, stroma, and thylakoids stacked in grana. The chlorophyll is built into the membrane of the thylakoids. Chlorophyll looks green because it absorbs the red and blue light, making these colors unavailable to be seen by your eyes. It is the green light which is not absorbed that finally reaches our eyes, making chlorophyll appear green. However, it is the energy from the red and blue light that are absorbed that is, thereby, able to be used to do photosynthesis. The green light we can see is not/cannot be absorbed by the plan, and thus cannot be used to do photosynthesis. What the main idea of paragraph 4 ? A. The plants can becolorfull in absorbing the photosynthesis process B. Photosynthesis needs absorbtion of colors C. The energy to do photosynthesis is provided not by the green light D. Chlorophyll enables photosynthesis to absorb colors E. Photosynthesis process gives colors to the plants
Soal 34	One day in July 2005, I went fishing on Sodus bay. I casted under our own dock with a rubber worm and Boom! A big bass took off to the other side of the dock and I was sure the line was going to break because his pulling was rubbing line againts sharp rocks. Three times I pulled him back to my side of the dock and three times he got got back to the other side of the dock. Finally, I got him to my side and he was tired. I knew that this was the biggest bass I had ever seen. I worked him towards me and got down and gripped him. However, I didn't have a good trip at first and I let him down to the water for a few seconds to get a better grip. The next thing I knew he was swimming away. No! Screamed louder than I ever had. It was the first time I cried over a fish. What a huge fish. This is a true story and sad one for me. The main idea of paragraph 2 is... A. The writer pulled the bass B. The writer got a big bass C. A big bass took off the sea D. The bass made the writer tired E. A big bass broke the writer's line

Soal 35	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Teks (1) The view that women are better parents than men has shown itself to be true throughout history. This is not to say that men are not of importance in child-rearing; indeed, they are most necessary if children are to appreciate fully the roles of both sexes. But women have proven themselves superior parents as a result of their conditioning, their less aggressive nature and generally better communication skills. (2) From the time they are little, females learn about nurturing. First with dolls and later perhaps with younger brothers and sisters, girls are given the role of career. Girls see their mothers in the same roles so it is natural that they identify this as female activity. Boy, in contrast, learn competitive roles far removed from what it means to nurture. While boy may dream of adventures, girls conditioning means they tend to see the future in terms of raising families. (3) Girls also appear to be less aggressive than boys. In adulthood, it is men not women, who prove to be the aggressors in crime and in war. Obviously, in raising children, a morepatient, gentle manner is preferable to a more aggressive one. Although, there certainly exist gentle men and aggressive women, by and large, female are less likely to resort to violence in attempting to solve problems. (4) Finally, women tend to be better communicator than men. This is shown in intelligence test, where females, on average, do better in verbal communication than males. Of course, communication is of utmost important in rearing children, as children tend to learn from and adopt the communication styles of their parents. (5) Thus, it is all very well to suggest a greater role for men in raising children, let us not forget that women are generally better suited to the parenting roles. As parents, women in general play a more important role than men because they are..... A. not aggressive at all B. good communicators C. superior human beings D. experienced in raising children E. capable of solving problems
Soal 36	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Teks (1) We all know that mobile phones, cell phones, hand-phones, whatever we want

	to call them (and shouldn't we all be calling them the same thing?) are changing our lives. But it takes a good old-fashioned survey to wake us up to the glaring reality: they have changed who we are. The mobile phone has indeed changed the way we behave. But perhaps we don't realize how much we have become its slave. Consider other elements of the Siemens Mobile Survey. With the exception of Australia, in every country surveyed the majority polled said they would go back for their phone if they left it at home (in Australia, it was a respectable 39%). If you've endured the traffic in Indonesia the Philippines and India, you'll know what kind of sacrifice some two-third of those surveyed are making. I can't think of anything I would go back for-except my wallet maybe, or my clothes. (2) And even if we remember to bring it, we're still not happy. Many of us get anxious if it hasn't rung or a text massage hasn't appeared for while (a while being about an hour). Once again of those surveyed, Indonesians (65%) and Filipinos (77%) get particularly jittery. Australians are more laid back about this (20%), but every other user in Asia seems to be glancing at the phone every few second. This statistic, I have to say is highly believable, and the isntict highly annoying. There's nothing worse than chatting to someone who constantly checks his or her hand-phone. (3) Then there's the fact that mobile phones are not only enslaving the user, they're ***trampling the right of everyone else***. Around a third of folk surveyed acknowledge they get so engrossed in mobile conversations that they're often unaware of speaking loudly while duscussing their private lives in public. At least most of us agree on one thing; with the exception of China, Hong Kong and Taiwan, the icreasing use of mobile phones has led to decline in courstesy and considerate behavior. (4) The bottom line here is that we are than a little bit out of contol. Mobile phones are great: But is we allow them to dominate our lives to this extent interrupting conversations with those around us, sending flirty text massages to random numbers – then I can only assume that in another 10 years, society as we know it will no longer exits. All we'll see is a blur of digital data going out and giving all the fun, socializing, falling in love and taking sneaky pictures of each other. Cell phones have not only enslaved the users but have also … A. Disturbed other people's conversation in public B. Eliminated the existence of social life C. Made the users behave impolitely D. Prevented people from socializing with each other E. Caused the user to become quite considerate
Soal 37	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Teks

	The government published a report yesterday saying that we need to eat more healthily more fruits and vegetables, less fat and sugar. So, that means fewer burgers, chips and fried foods as well as cutting down on sweet things. We went into Central London yesterday at lunch time and asked people what they thought about it. "It's got nothing to do with government what I eat," said Paul Keel, a building worker, and Dorothi Mattews, a teacher as they ate beef burgers and chips washed down with strawberry milkshakes. Tim Kennnor, however, welcome the government's advice. Then, we asked the Simpson's restaurant's manager about it. "I don't think people believe all these reports anymore. What they say is good for you on June, then they say bad for you on July. People stopped taking notice. We serve what we've always served. Almost all of it is fattening." The restaurant manager think of the report that … A. People don't understand the advice given. B. The government's advice is inconsistent. C. People are more concerned about losing weight. D. Fattening foods are more delicious. E. People will welcome the government advice.
Soal 38	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Teks (1) The killer sea waves known as Tsunamis are so quiet in their approach from a far, so harmless, that until recently their history has been one of surprise attack. (2) Out in the middle of the ocean, the distance between tsunami wave crests can be 100 miles and the height of the wave have no more than three feet: sailors can ride one and suspect nothing. At the shoreline, the first sign is often an ebbing-a retreat but rather a gathering of forces. When the great waves finally do strike, they rear up and hit harbor and coast, causing death and damage. (3) These seismic sea waves or tidal waves-as they are sometimes called, bear no relation to the moon or tides. And the word "tsunami", Japanese for "harbor wave", related to their destination rather than their origin. The causes are various: undersea or coastal earthquakes, deep ocean avalanches or volcanism. Whatever the cause, the wave motion starts with a sudden move like a hit from a giant paddle that displaces the water. And the greater the undersea hit, the greater the tsunami's damaging power. (4) In 1883, Krakatoa volcano in the East Indies erupted, and the entire island collapsed in 820 feet of water. A tsunami is a tremendous forces bouncing around

	Java and Sumatera, killing 36.000 people with walls of water that reached 115 feet in height. (5) In 1946, a tsunami struck first near Alaska and then, without warning, hit Hawaiian Islands, killing 159 people and causing millions of dollars of damage. This led to the creation of the Tsunami Warning System, whose nerve center in Honolulu keeps around-the-clock vigil with the aid of new technology. If the seismic sea waves are confirmed by the Honolulu center, warnings are transmitted within a few hours to all threatened Pacific points. Tsunamis have been deprived of their most deadly sting-surprise. Which of the recorded tsunami in the text was the worst? The one that hit ..... A. The East indies B. The Island of Java C. Alaska D. The Hawaiian Islands E. Honolulu
Soal 39	The MV AcademicShokalskiy, a "highly ice-strengthened"Russian tour ship built in Finland in 1984 "for polar and oceanographic research" is stranded in Antartica's summer ice with 74 passengers and crew members aboard. The group, which includes two Guardian journalist is retracing the harrowing 1911 Antarctic expedition led by Sir Douglas Mawson, who lost many of his team members and nearly died himself on the frigid continent a century ago. The ship's passengers include an Australian research team led by University of New South Wales Professor Chris Turney, who said in November that the voluminous data collected by Mawson 100 years ago is critical to understanding global warming. But Turney reported that blizzard-like conditions and thick ocean ice are preventing the latest expedition from leaving "Unfortunatly proceeding north we found out path blocked by ice pushed in by an increasingly strong southeasterly wind. On Christmas eve realized we could not get through, in spite of being just 2 nautical miles from open water." Turney report in his blog. "According to reports nobody is in present danger and there nearby icebreakers are being sent to assis", sain Expeditions online.com, which books polar expeditions. The ship is stuck par-way through her Australian Antarctic Expedition towards Mawson's Hut at Cape Denison. Located about 100 nautical miles east of Dumoni D'unville a French base on Antartica, and 1500 nautical miles south of Hobart in Tasmania. Three icebreakers-China's Xue Long. Australia's Aurora Australis, and France's L'Astrolabe-have been dispatched to the Australian maritime Safety Authority (AMSA), which is coordinating the international rescue after the Falmouth Maritime Rescue Coordinator Centre in the United Kingdom received a satellite distress call Christmas morning. However, it will take the icebreakers at least two

	days to get to the stranded ship, which is experiencing very strong winds and limited visibility. The closest rescue ship is not expected to get to the scane until sometimes Friday nght. "Which it is early winter in the Arctic, it is early summer in the Antarctic. Continuing patterns seen in recnt years. Antarctic sea ice extent remain unusually high, near of above previous daily maximum values," according to the National Snow and ice Data Center (NSIDC) The purpose of the text is to… A. Report about a stranded ship B. Describe efforts to rescue a stranded ship C. Report the lost of Russian ship D. Explain why the Russian ship was stranded E. Discuss problems of rescuing a stranded
Soal 40	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Teks 3 (1) An intense heat wave is baking Europe this week, with countries from Spain to England setting record high temperatures. The hot days and warm nights are creating dangerous conditions for those unable to escape the heat, AccuWeather warned. The heat shows no signs of letting up. "We have a lot of heat-wave days ahead of us," MeteoFrance forecaster Francois Gourand told the Associated Press, noting a wide swath of southern France from Toulouse to Lyon was facing temperatures up to 105 degrees through the middle of next week. Germany and Poland will begin to feel the worst of the heat wave Thursday, AccuWeather said. (2) The high temperatures in Europe -following blistering, deadly heat waves in India and Pakistan that killed thousands in May and June-are helping push 2015's global temperatures to the highest in recorded human history, according to the Weather Underground. Europeans are wary of heat waves, with the murderous heat of 2003 when 70,000 died still fresh in people's minds. In France, where 20,000 died in 2003, a nationwide heat emergency plan has been enacted this week, the Guardian reported. This includes opening cooling centers and checking on the elderly and vulnerable. (3) An all-time record high temperature for the United Kingdom in July was set Wednesday at London's Heathrow Airport, as the mercury soared to 98.6 degrees. Wimbledon's tennis matches in London were being competed on the hottest day in the tournament's history, which goes back to 1887. A ball boy collapsed in the heat during a match Wednesday and needed to be taken off the court on stretcher, the AP reported. Some commuters outside a London subway weren't bothered by the sweltering heat. " I'm loving it. I can't complain," said Maggie Cloud, a university students who planned to relax in the park. "We pay so much money to

	go abroad to holidays, and now we have the weather here. It's cheaper," she told the AP. Temperatures climed to 104 degrees in n Madrid on Monday, breaking the city's all- time June high at Gatafe Airport. (4) The heat is the result of an unusual bulge in the jet stream, which is allowing a massive ridge of high pressure to surge over the continent, bringing in extremely hot air from North Africa. Even beyond the next few days, the weather pattern across much of Europe will feature warmer-than- normal temperatures through the middle of July with only a few brief breaks from the warmth, said AccuWeather meteorologist Adam Douty. The purpose of the text is … A. To discuss a plan to relax in the rarely weather. B. To remind people in Europe that heat wave killed people. C. To tell what happen to people during heat wave condition D. To review heat wave disaster across the country. E. To describe heat wave attack's in Europe.
Soal 41	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text Colors, hair, and jewelry are frequently determined by a person's sex. This is not always true for all cultures, and it is not even true now throughout the United States. In this country, there were protests to bring about a change from these culturally strict norms. The 'anti-military' attitudes of the 1960s and the 1970s sought to break with the military tradition masculine or 'macho' position, thus making it more acceptable for men to wear floral designs on their shirts in pinks, purples, violets, and other 'feminine colors'. For some people, long hair and jewelry on men also became acceptable as means of expressing this changed way of thinking. More facial hair also became common, precisely because it differed from the military norm. The business world, however, has been slow to change. The topic of the text is about … A. The change in gender-based attitude in the U.S B. Jewelry as a means of expressing thoughts C. Feminine colors versus masculine colors D. The anti military movement in the U.S. E. Various cultures in the world

Soal 42

Read the following the text, then choose the correct answer by crossing A, B, C, D OR E!

Text

(1) An intense heat wave is baking Europe this week, with countries from Spain to England setting record high temperatures. The hot days and warm nights are creating dangerous conditions for those unable to escape the heat, AccuWeather warned. The heat shows no signs of letting up. “We have a lot of heat-wave days ahead of us,” MeteoFrance forecaster Francois Gourand told the Associated Press, noting a wide swath of southern France from Toulouse to Lyon was facing temperatures up to 105 degrees through the middle of next week. Germany and Poland will begin to feel the worst of the heat wave Thursday, AccuWeather said.

(2) The high temperatures in Europe -following blistering, deadly heat waves in India and Pakistan that killed thousands in May and June-are helping push 2015's global temperatures to the highest in recorded human history, according to the Weather Underground. Europeans are wary of heat waves, with the murderous heat of 2003 when 70,000 died still fresh in people's minds. In France, where 20,000 died in 2003, a nationwide heat emergency plan has been enacted this week, the Guardian reported. This includes opening cooling centers and checking on the elderly and vulnerable.

(3) An all-time record high temperature for the United Kingdom in July was set Wednesday at London's Heathrow Airport, as the mercury soared to 98.6 degrees. Wimbledon’s tennis matches in London were being competed on the hottest day in the tournament’s history, which goes back to 1887. A ball boy collapsed in the heat during a match Wednesday and needed to be taken off the court on stretcher, the AP reported. Some commuters outside a London subway weren’t bothered by the sweltering heat. “ I’m loving it. I can’t complain,” said Maggie Cloud, a university students who planned to relax in the park. “We pay so much money to go abroad to holidays, and now we have the weather here. It’s cheaper,” she told the AP. Temperatures climed to 104 degrees in n Madrid on Monday, breaking the city's all- time June high at Gatafe Airport.

(4) The heat is the result of an unusual bulge in the jet stream, which is allowing a massive ridge of high pressure to surge over the continent, bringing in extremely hot air from North Africa. Even beyond the next few days, the weather pattern across much of Europe will feature warmer-than- normal temperatures through the middle of July with only a few brief breaks from the warmth, said AccuWeather meteorologist Adam Douty.

The topic of the text above is …

A. Cautions of heat wave.

B. Blistering heat waves in Asia.

C. Dangerous heat wave in Europe.

D. Warning from a meteorologist.

E. Highest global temperatures in human history

Soal 43

Read the following the text, then choose the correct answer by crossing A, B, C, D OR E!

Text

(1) We often think it very funny if a film shows a woman or a girl screaming and running in fright at the sight of a mouse or a cockroach. But we don't consider our own little, secret fers a laughing matter. Perhaps you shiver at the sight of a snake or an eel, cannot bear to hear the sound of jagged metal rubbing against metal, or feel uneasy in a lift. Fortunately, most of us suffer only mild from of fright about certain things. According to psychiatrists, this is quite normal. It becomes a problem only when the fear develops into a mental condition that affects all aspects of a person's life. Studies of thousands of people with an abnormal dread show that every time they think about- or come into contact with – the subject of their fear, they show all the physical signs of a person who is facing serious danger: a racing heart, dry mouth, could sweat and weak knees.

(2) Doctors and psychiatrists welcome publicity about phobias, for this helps to convince their patients that their petients that their patients that their fears are not unique. It reassures the patients to know that there are other people who are afraid of the same thing, and it makes the patients more willing to try to understand what is causing their fear. Once they realize what in their subconscious is causing their fear, they are half-way to being cured. Psychiatrists believe that these unreasonable fears are caused by deep-seated reasons, perhaps a terrifying experience during childhood.

(3) One typical case was Mrs. Marry Batchelor, a London housewife, who suffered from agoraphobia (fear of open spaces). For twenty-three years, she remains indor. Following publicity about other agoraphobia cases, she was persuaded to go out for the first time to watch her son perform with pop group.

(4) Most of use suffer from claustrophobia (fear of being in closed spaces) iin a mild way, but the real sufferers go about in great fear of being trapped without escape in a lift, in packed train or in an aircraft. Altogether, there are a hundread and thirty phobias listed in the medical dictionary, ranging from acrophobia (fear of heights) to xenophobia (morbid dislike of foreigners). Some phobias are very odd indeed. There are cases of people who turn cold with terror at the sight of a cabbage leaf, run away in fright at the sight of a brid or fraint when they hear hear a dog bark. There was even a man who refused to eat anything that he knew had been touched by someone who had handled matches.

In which of the following combined courses would this passage probably be used as assigned reading …

A. Psychology / sociology

B. Medicine/ biology

C. Philosophy/ medicine

D. Psychology/medicine

E. Sociology/biology

Soal 44	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text (1) BAGHDAD (Reteurs) – Six U.S. soldiers were killed by a blast north of Bagdad on Monday, the U.S. military said on Tuesday, in one of the single deadliest ground attacks against American forces in Iraq in months. (2) Three other soldiers were wounded after an explosion went off near their vehicles in the province of Salahaddin, the military said in a statement without giving details. Salahaddin is a Sumi, Arab insurgent stronghold. (3) More than 3,170 U.S. soldiers have died in Iraq since the U.S. led invention in March 2003. Iraqi and U.S. forces are in third week of security crackdown in Baghdad aimed at stemming sectarian violence. U.S. military commanders have expressed concern that insurgents could step up attacks outside Baghdad. The word **crackdown** in Paragraph 3 means A. Line B. Guard C. Attack D. Clean-up E. Protection
Soal 45	Read the following the text, then choose the correct answer by crossing A, B, C, D OR E! Text (1) An intense heat wave is baking Europe this week, with countries from Spain to England setting record high temperatures. The hot days and warm nights are creating dangerous conditions for those unable to escape the heat, AccuWeather warned. The heat shows no signs of letting up. “We have a lot of heat-wave days ahead of us,” MeteoFrance forecaster Francois Gourand told the Associated Press, noting a wide swath of southern France from Toulouse to Lyon was facing temperatures up to 105 degrees through the middle of next week. Germany and Poland will begin to feel the worst of the heat wave Thursday, AccuWeather said. (2) The high temperatures in Europe -following blistering, deadly heat waves in India and Pakistan that killed thousands in May and June-are helping push 2015's global temperatures to the highest in recorded human history, according to the Weather Underground. Europeans are wary of heat waves, with the murderous heat of 2003 when 70,000 died still fresh in people's minds. In France, where 20,000 died in 2003, a nationwide heat emergency plan has been enacted this week, the Guardian reported. This includes opening cooling centers and checking

	on the elderly and vulnerable. (3) An all-time record high temperature for the United Kingdom in July was set Wednesday at London's Heathrow Airport, as the mercury soared to 98.6 degrees. Wimbledon's tennis matches in London were being competed on the hottest day in the tournament's history, which goes back to 1887. A ball boy collapsed in the heat during a match Wednesday and needed to be taken off the court on stretcher, the AP reported. Some commuters outside a London subway weren't bothered by the sweltering heat. " I'm loving it. I can't complain," said Maggie Cloud, a university students who planned to relax in the park. "We pay so much money to go abroad to holidays, and now we have the weather here. It's cheaper," she told the AP. Temperatures climed to 104 degrees in n Madrid on Monday, breaking the city's all- time June high at Gatafe Airport. (4) The heat is the result of an unusual bulge in the jet stream, which is allowing a massive ridge of high pressure to surge over the continent, bringing in extremely hot air from North Africa. Even beyond the next few days, the weather pattern across much of Europe will feature warmer-than- normal temperatures through the middle of July with only a few brief breaks from the warmth, said AccuWeather meteorologist Adam Douty. The following options is closest in meaning to the word "baking" (paragraph 1) is … A. scorching B. waving C. drying D. making E. drought